A Measure-Theoretic Foundation for Meta-Analysis

Pooling, Heterogeneity, and Epistemic Limits under the Half-Incorrectness Framework

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Abstract

Meta-analysis is the statistical procedure of pooling effect estimates across a collection of independent studies in order to obtain a combined inference. As a methodology, it is itself a meta-theory in the precise sense established by Ghosh (2026): its domain consists of theories (primary studies), and it imposes evaluative standards upon them. This paper develops a rigorous measure-theoretic treatment of meta-analysis grounded in the Half-Incorrectness framework. We formalise the meta-analytic study space as a measurable space of admissible primary studies, define a family of pooling operators, and embed the classical fixed-effect and random-effects models within the abstract framework. We prove that the Half-Incorrectness Axiom ($\mu(\mathcal{I}_E) \geq \frac{1}{2}$) implies a structural lower bound on the expected pooling error under any weighting scheme, provided the incorrectness of a study is identified with the non-negligibility of its bias. We formalise heterogeneity as a measure-theoretic dispersion property of the study space and show that the presence of heterogeneity above a critical threshold makes universal pooling incoherent in the sense of Ghosh (2026). The meta-analytic protocol is itself an admissible meta-theory, and under reflexive inclusion it cannot certify universal correctness of its own aggregations. We introduce the *Epistemic Weight Function* as a rigorous replacement for the informal notion of study quality, establish conditions under which weighted pooling is asymptotically consistent despite population-level half-incorrectness, and correct a common misinterpretation of the entropy of pooled estimates. Connections are drawn to publication bias, Bayesian hierarchical models, network meta-analysis, and the philosophy of evidence synthesis.

The paper ends with "The End"

1 Introduction

Meta-analysis occupies a peculiar epistemic position. On one hand it is the most highly regarded tool in evidence-based medicine, educational research, psychology, economics, and public policy: a systematic aggregation of evidence that, in principle, cancels idiosyncratic noise and converges on a true effect. On the other hand, meta-analyses have been criticised for pooling heterogeneous studies, amplifying publication bias, and producing false precision [15, 12].

The companion paper by Ghosh (2026) provides a formal framework within which this tension can be stated precisely. A meta-theory is a theory whose domain consists of theories. A meta-analysis is exactly that: it takes a collection of primary studies (theories, in the generalised sense) as its domain and imposes a correctness criterion upon them through effect pooling. The central axiom of Ghosh (2026)—the *Half-Incorrectness Axiom*—states that when a probability measure μ is placed over the admissible space $\mathcal{M}_{\mathcal{L}}$ of meta-theories, the measure of incorrect ones satisfies $\mu(\mathcal{I}_E) \geq \frac{1}{2}$.

The present paper asks what this axiom implies for meta-analysis specifically. We proceed in four broad movements:

(i) We embed meta-analysis within the Ghosh (2026) framework by formalising the study space as a measurable space, defining study-level correctness, and identifying the meta-analytic protocol itself as an admissible meta-theory.

(ii) We prove that half-incorrectness of the study population implies a non-trivial lower bound on pooling error, and that no weighting scheme eliminates this bound without auxiliary information.

(iii) We formalise heterogeneity as a measure-theoretic property and show that when heterogeneity exceeds a critical threshold, pooling becomes incoherent under the framework.

(iv) We develop the Bayesian and entropic consequences, drawing connections to publication bias, network meta-analysis, and model-averaging.

Throughout, we adhere to the convention of Ghosh (2026): correctness is always *regime-relative*, the space of admissible objects is always *language-relative*, and reflexive participation is always *possible* without necessarily generating contradiction.

2 Foundations: The Study Space and Its Measure

2.1 Primary Studies as Theories

A primary study is an empirical or theoretical report that estimates a quantity of interest—a causal effect, a prevalence, a coefficient—under a set of design and modelling assumptions. In the generalised sense of Definition 1 of Ghosh (2026), a primary study is a theory: it generates predictions or classifications and is subject to evaluation.

Definition 2.1 (Primary Study)**.** A *primary study* is a tuple

$$s = (\hat{\theta}_s,\, \sigma_s^2,\, \Pi_s,\, \Delta_s),$$

where $\hat{\theta}_s \in \mathbb{R}$ is the reported effect estimate, $\sigma_s^2 > 0$ is the within-study variance, Π_s is the study protocol (design, population, intervention, outcome), and Δ_s is the set of modelling and inferential assumptions invoked.

Definition 2.2 (Study Space)**.** Let $\mathcal{L}$ be a language in which study protocols and assumptions can be encoded. The *admissible study space* is

$$\mathcal{S}_{\mathcal{L}} = \{s : s \text{ is a primary study encodable in } \mathcal{L}\}.$$

2.2 The Meta-Analytic Evaluation Regime

Following Definition 5 of Ghosh (2026), we specify an evaluation regime for primary studies.

Definition 2.3 (Meta-Analytic Evaluation Regime)**.** A *meta-analytic evaluation regime* is a tuple

$$E_{\mathrm{MA}} = (D,\, S,\, \kappa_{\mathrm{MA}}),$$

where D is the phenomenon of interest (the causal or associational quantity being estimated), S is a semantic standard specifying how effect estimates are to be interpreted, and κ_{MA} is the correctness criterion:

$$\kappa_{\mathrm{MA}}(s) = 1 \iff \mathrm{Bias}(\hat{\theta}_s) \leq \varepsilon \text{ and } \Pi_s \text{ is internally valid for } D,$$

for a pre-specified tolerance $\varepsilon \geq 0$.

Remark 2.4. The tolerance ε encodes the degree of bias that an evaluation regime is willing to absorb. Setting $\varepsilon = 0$ corresponds to a strict regime; practical regimes allow $\varepsilon > 0$ to accommodate small design imperfections.

2.3 Measure on the Study Space

Assumption 2.5 (Measurable Study Space)**.** The admissible study space is equipped with a σ-algebra $(\mathcal{S}_{\mathcal{L}},\, \mathcal{A})$.

Assumption 2.6 (Study Measure)**.** There exists a probability measure $\nu : \mathcal{A} \to [0,1]$ such that $\nu(\mathcal{S}_{\mathcal{L}}) = 1$. The measure ν may represent the empirical distribution of published studies, a prior over all conceivable studies, or a complexity-weighted distribution over study protocols.

Assumption 2.7 (Measurable Incorrectness)**.** The incorrectness set $\mathcal{I}_E^{\mathrm{MA}} = \{s \in \mathcal{S}_{\mathcal{L}} : \kappa_{\mathrm{MA}}(s) = 0\}$ is measurable: $\mathcal{I}_E^{\mathrm{MA}} \in \mathcal{A}$.

Axiom 2.8 (Half-Incorrectness of the Study Population)**.** Relative to $(\mathcal{S}_{\mathcal{L}},\, \mathcal{A},\, \nu,\, E_{\mathrm{MA}})$,

$$\nu\big(\mathcal{I}_E^{\mathrm{MA}}\big) \geq \tfrac{1}{2}.$$

This is the direct instantiation of Axiom 1 of Ghosh (2026) to the meta-analytic setting. Its empirical motivation is substantial: reviews of registered trials, replication studies in psychology and medicine, and assessments of epidemiological cohort studies consistently find that the majority of primary studies contain one or more sources of non-trivial bias [4, 14, 19].

3 Pooling Operators and Classical Models

3.1 Abstract Pooling

A meta-analytic pooling operator aggregates a finite collection of study estimates into a single summary estimate.

Definition 3.1 (Pooling Operator)**.** Let $\mathbf{s} = (s_1, \ldots, s_k)$ be a finite ordered sample of primary studies drawn from $\mathcal{S}_{\mathcal{L}}$. A *pooling operator* is a measurable map

$$\Phi : \mathcal{S}_{\mathcal{L}}^k \to \mathbb{R}$$

that returns a combined effect estimate $\hat{\theta}_\Phi = \Phi(\hat{\theta}_{s_1}, \ldots, \hat{\theta}_{s_k})$.

Definition 3.2 (Weighted Linear Pooling Operator)**.** A pooling operator is a *weighted linear pooling operator* if

$$\Phi_w(\mathbf{s}) = \sum_{i=1}^k w_i \hat{\theta}_{s_i}, \qquad w_i \geq 0, \quad \sum_{i=1}^k w_i = 1.$$

3.2 Fixed-Effect Model

In the fixed-effect (common-effect) model, all studies are assumed to estimate a single underlying effect θ_0.

Definition 3.3 (Fixed-Effect Model)**.** Under the fixed-effect model, for each study s_i,

$$\hat{\theta}_{s_i} = \theta_0 + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_{s_i}^2), \quad \varepsilon_i \text{ independent.}$$

The fixed-effect pooled estimate is

$$\hat{\theta}_{\mathrm{FE}} = \frac{\sum_{i=1}^k w_i^{\mathrm{FE}} \hat{\theta}_{s_i}}{\sum_{i=1}^k w_i^{\mathrm{FE}}}, \qquad w_i^{\mathrm{FE}} = \frac{1}{\sigma_{s_i}^2}.$$

3.3 Random-Effects Model

The random-effects model acknowledges that the true effects may differ across studies.

Definition 3.4 (Random-Effects Model)**.** Under the random-effects model,

$$\hat{\theta}_{s_i} = \theta_i + \varepsilon_i, \qquad \theta_i = \theta_0 + \delta_i, \qquad \delta_i \sim \mathcal{N}(0, \tau^2), \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_{s_i}^2),$$

where $\tau^2 \geq 0$ is the *between-study variance* (heterogeneity parameter). The DerSimonian–Laird pooled estimate uses weights

$$w_i^{\mathrm{RE}} = \frac{1}{\sigma_{s_i}^2 + \hat{\tau}^2},$$

where $\hat{\tau}^2$ is an estimate of τ^2.

Remark 3.5. The fixed-effect model is a special case with $\tau^2 = 0$. The random-effects model is strictly more general and is the appropriate default when heterogeneity is suspected.

4 Pooling Error under Half-Incorrectness

4.1 Bias Decomposition

We decompose the pooled estimate into a contribution from correct and incorrect studies.

Definition 4.1 (Study Bias Function)**.** For a study $s \in \mathcal{S}_{\mathcal{L}}$, define the *bias function*

$$b(s) = \mathbb{E}[\hat{\theta}_s] - \theta_0,$$

where θ_0 is the true effect under the evaluation regime E_{MA}. The study is in $\mathcal{I}_E^{\mathrm{MA}}$ if and only if $|b(s)| > \varepsilon$.

Definition 4.2 (Aggregate Pooling Bias)**.** For a weighted linear pooling operator Φ_w applied to a sample $\mathbf{s} = (s_1, \ldots, s_k)$,

$$\mathrm{Bias}(\hat{\theta}_{\Phi_w}) = \mathbb{E}[\hat{\theta}_{\Phi_w}] - \theta_0 = \sum_{i=1}^k w_i \, b(s_i).$$

4.2 Lower Bound on Expected Pooling Error

Theorem 4.3 (Half-Incorrectness Pooling Bound)**.** *Let Axiom 2.8 hold and let* $\mathbf{s} = (s_1, \ldots, s_k)$ *be an i.i.d. sample from* ν*. Let* Φ_w *be any weighted linear pooling operator with equal weights* $w_i = 1/k$*, and let* $B > \varepsilon$ *be such that* $|b(s)| \geq B$ *for all* $s \in \mathcal{I}_E^{\mathrm{MA}}$*. Then the expected absolute pooling bias satisfies*

$$\mathbb{E}_{\nu^k}\left[\left|\mathrm{Bias}(\hat{\theta}_{\Phi_w})\right|\right] \geq \frac{B-\varepsilon}{2} \cdot \left(1 - e^{-k/4}\right).$$

In particular, as $k \to \infty$*,*

$$\liminf_{k\to\infty} \mathbb{E}_{\nu^k}\left[\left|\mathrm{Bias}(\hat{\theta}_{\Phi_w})\right|\right] \geq \frac{B-\varepsilon}{2}.$$

Proof. Let $Z_i = \mathbf{1}[s_i \in \mathcal{I}_E^{\mathrm{MA}}]$ for $i = 1, \ldots, k$. By Axiom 2.8, $\mathbb{P}(Z_i = 1) \geq \frac{1}{2}$. Define $\bar{Z}_k = \frac{1}{k}\sum_{i=1}^{k} Z_i$. Then

$$\mathrm{Bias}(\hat{\theta}_{\Phi_w}) = \frac{1}{k}\sum_{i=1}^{k} b(s_i) = \frac{1}{k}\sum_{i:Z_i=1} b(s_i) + \frac{1}{k}\sum_{i:Z_i=0} b(s_i).$$

For $s \in \mathcal{I}_E^{\mathrm{MA}}$, $|b(s)| \geq B$. For $s \notin \mathcal{I}_E^{\mathrm{MA}}$, $|b(s)| \leq \varepsilon$. By the triangle inequality,

$$\left|\mathrm{Bias}(\hat{\theta}_{\Phi_w})\right| \geq B\,\bar{Z}_k - \varepsilon(1 - \bar{Z}_k) = (B+\varepsilon)\bar{Z}_k - \varepsilon.$$

Taking expectations and applying Hoeffding's inequality to bound $\mathbb{P}(\bar{Z}_k < \frac{1}{4})$ from above by $e^{-k/4}$,

$$\mathbb{E}\left[\bar{Z}_k\right] \cdot (B+\varepsilon) - \varepsilon \geq \tfrac{1}{2}(B+\varepsilon) - \varepsilon = \frac{B-\varepsilon}{2}.$$

The exponential correction accounts for the probability mass on the event $\bar{Z}_k < \frac{1}{4}$, which is bounded by $e^{-k/4}$ and vanishes as $k \to \infty$. □

Corollary 4.4 (No Uniformly Unbiased Pooling)**.** *Under the conditions of Theorem 4.3, no uniformly weighted pooling operator is asymptotically unbiased unless* $B \leq \varepsilon$*.*

Remark 4.5. The bound $\frac{B-\varepsilon}{2}$ is tight in the sense that it is achieved when exactly half the studies have bias B and the remaining half have bias 0. This corresponds to the boundary case $\nu(\mathcal{I}_E^{\mathrm{MA}}) = \frac{1}{2}$ of Axiom 2.8.

4.3 Variance Decomposition under Random Effects

Proposition 4.6 (Variance Decomposition)**.** *Under the random-effects model (Definition 3.4), the mean squared error of the pooled estimate decomposes as*

$$\mathrm{MSE}(\hat{\theta}_{\Phi_w}) = \underbrace{\left[\mathrm{Bias}(\hat{\theta}_{\Phi_w})\right]^2}_{bias^2} + \underbrace{\sum_{i=1}^{k} w_i^2(\sigma_{s_i}^2 + \tau^2)}_{variance}.$$

When Axiom 2.8 holds and $B > \varepsilon$*, the bias term is bounded below by* $\left(\frac{B-\varepsilon}{2}\right)^2$ *in expectation over* ν^k*.*

Proof. The variance decomposition follows from the standard bias-variance identity $\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Var}$. The lower bound on the bias term follows from Theorem 4.3 via Jensen's inequality applied to the square function. □

5 Heterogeneity as a Measure-Theoretic Property

5.1 Formalising Heterogeneity

Classical meta-analysis operationalises heterogeneity through the I^2 statistic or the Q-test [11]. We provide a measure-theoretic counterpart that connects naturally to the Ghosh (2026) framework.

Definition 5.1 (Effect Distribution)**.** The *effect distribution* induced by the study measure ν is the push-forward measure

$$\rho = \nu \circ \theta^{-1},$$

where $\theta : \mathcal{S}_{\mathcal{L}} \to \mathbb{R}$ maps each study to its true underlying effect θ_s.

Definition 5.2 (Measure-Theoretic Heterogeneity)**.** The *measure-theoretic heterogeneity* of the study population is

$$\tau_\nu^2 = \int_{\mathbb{R}} (\theta - \bar{\theta})^2 \, d\rho(\theta),$$

where $\bar{\theta} = \int_{\mathbb{R}} \theta \, d\rho(\theta)$ is the population mean effect. The study population is *homogeneous* if $\tau_\nu^2 = 0$ and *heterogeneous* if $\tau_\nu^2 > 0$.

5.2 Heterogeneity and Incoherence of Universal Pooling

Theorem 5.3 (Heterogeneity Incoherence Threshold)**.** *Let $\tau_\nu^2 > 0$ and let Φ_w be a weighted linear pooling operator. Then for any target $\theta_0 \in \mathbb{R}$,*

$$\mathbb{E}_\nu\left[(\theta_s - \theta_0)^2\right] = (\bar{\theta} - \theta_0)^2 + \tau_\nu^2.$$

In particular, no choice of θ_0 simultaneously minimises $(\theta_s - \theta_0)^2$ for ν-almost-all studies. When τ_ν^2 exceeds the pooled within-study variance $\bar{\sigma}^2 = \int \sigma_s^2 \, d\nu(s)$, the effective pooling precision is dominated by between-study variation, and the pooled estimate $\hat{\theta}_{\Phi_w}$ provides no meaningful precision gain over any single study.

Proof. The decomposition follows from the identity $\mathbb{E}[(X - c)^2] = (\mathbb{E}[X] - c)^2 + \mathrm{Var}(X)$ applied to the random variable $\theta_s \sim \rho$ with $c = \theta_0$. The incoherence claim follows because the integrand $(\theta_s - \theta_0)^2$ cannot be simultaneously zero for all s in the support of ν when $\tau_\nu^2 > 0$. The precision claim follows from the inequality $\mathrm{Var}(\hat{\theta}_{\Phi_w}) \geq \tau_\nu^2/k$ for equal weights, which exceeds $\mathrm{Var}(\hat{\theta}_{s_i}) = \sigma_{s_i}^2$ when $\tau_\nu^2 > k\,\bar{\sigma}^2$. ☐

Corollary 5.4 (Incoherence of Universal Correctness under Heterogeneity)**.** *Suppose the study population is heterogeneous ($\tau_\nu^2 > 0$) and Axiom 2.8 holds. Then no meta-analytic protocol can correctly pool all studies in $\mathcal{S}_{\mathcal{L}}$ and simultaneously be correct relative to E_{MA} for every sub-population with distinct effect θ_s.*

Remark 5.5. Corollary 5.4 is the meta-analytic analogue of Corollary 2 of Ghosh (2026): no global certifier can correctly certify universal correctness. In the present setting, universal pooling acts as a global certifier, and heterogeneity is the structural obstacle.

6 The Meta-Analytic Protocol as a Meta-Theory

6.1 Reflexive Participation of Meta-Analysis

Let T_{MA} denote the meta-theory whose central assertion is a specific meta-analytic protocol Φ_w. Following Section 7 of Ghosh (2026), we ask whether T_{MA} belongs to its own evaluative domain.

Assumption 6.1 (Reflexive Inclusion of Meta-Analysis)**.** The meta-analytic protocol is itself an admissible theory:

$$T_{\mathrm{MA}} \in \mathcal{M}_{\mathcal{L}}.$$

Theorem 6.2 (Reflexive Non-Exemption of Meta-Analysis)**.** *If Assumption 6.1 holds and the incorrectness predicate $I_E(\cdot)$ is defined on $\mathcal{M}_{\mathcal{L}}$, then $I_E(T_{\mathrm{MA}}) \in \{0, 1\}$. No meta-analytic protocol can exempt itself from evaluation without either exiting $\mathcal{M}_{\mathcal{L}}$ or restricting the domain of I_E.*

Proof. This follows directly from Theorem 2 of Ghosh (2026) with T_{HI} replaced by T_{MA}. The incorrectness predicate $I_E : \mathcal{M}_{\mathcal{L}} \to \{0, 1\}$ is defined on all of $\mathcal{M}_{\mathcal{L}}$; since $T_{\mathrm{MA}} \in \mathcal{M}_{\mathcal{L}}$, the value $I_E(T_{\mathrm{MA}})$ is well-defined. ☐

6.2 Self-Assessment and Coherent Self-Failure

A meta-analysis can fail in several ways: by incorrectly pooling heterogeneous studies, by inadequately adjusting for publication bias, or by applying a fixed-effect model where random-effects is warranted.

Proposition 6.3 (Coherent Self-Failure of Meta-Analysis)**.** *It is coherent for a meta-analytic protocol T_{MA} to be incorrect relative to E_{MA} without generating a logical contradiction. Specifically, if T_{MA} produces a pooled estimate $\hat{\theta}_{\Phi_w}$ for which $\mathrm{Bias}(\hat{\theta}_{\Phi_w}) > \varepsilon$, then $I_{E_{\mathrm{MA}}}(T_{\mathrm{MA}}) = 1$, and this truth value is semantically well-defined.*

Proof. By Proposition 2 of Ghosh (2026), self-reference does not automatically produce inconsistency. The truth value of $I_{E_{\mathrm{MA}}}(T_{\mathrm{MA}}) = 1$ is defined by whether the pooled estimate satisfies the correctness criterion κ_{MA}, which is an objective property of $\hat{\theta}_{\Phi_w}$ and θ_0. No Liar-type paradox arises because $I_E(T_{\mathrm{MA}})$ is determined by the model, not by T_{MA}'s own assertion. □

7 The Epistemic Weight Function

7.1 Motivation

The classical inverse-variance weights $w_i = 1/\sigma^2_{s_i}$ optimise pooled variance under the assumption of correctness. Under the Half-Incorrectness Axiom, this is insufficient: a study may have low within-study variance but high bias. We introduce the *Epistemic Weight Function* as a rigorous replacement for informal notions of study quality.

Definition 7.1 (Epistemic Weight Function)**.** An *Epistemic Weight Function* is a measurable map

$$\omega : \mathcal{S}_{\mathcal{L}} \to [0, \infty)$$

such that:

(i) $\omega(s) > 0$ only if $s \notin \mathcal{I}_E^{\mathrm{MA}}$ (incorrect studies receive zero weight), or more generally, $\omega(s)$ is decreasing in $|b(s)|$;

(ii) $\int \omega(s)\, d\nu(s) = 1$;

(iii) $\omega(s) \geq w_s^{\mathrm{FE}}$ whenever s is correct and has smaller within-study variance than the population average.

Theorem 7.2 (Consistency under Epistemic Weighting)**.** *Suppose the Epistemic Weight Function ω assigns zero weight to all incorrect studies: $\omega(s) = 0$ for all $s \in \mathcal{I}_E^{\mathrm{MA}}$. Let*

$$\hat{\theta}_{\omega,k} = \frac{\sum_{i=1}^{k} \omega(s_i)\, \hat{\theta}_{s_i}}{\sum_{i=1}^{k} \omega(s_i)}.$$

Then $\hat{\theta}_{\omega,k} \xrightarrow{p} \theta_0$ as $k \to \infty$, provided that the correct studies are consistent estimators of θ_0 in the evaluation regime E_{MA}.

Proof. When $\omega(s) = 0$ for $s \in \mathcal{I}_E^{\mathrm{MA}}$, the weighted sum effectively restricts to $\mathcal{S}_{\mathcal{L}} \setminus \mathcal{I}_E^{\mathrm{MA}}$. By assumption, $\nu(\mathcal{I}_E^{\mathrm{MA}}) \leq 1$, so the correct study set has measure $1 - \nu(\mathcal{I}_E^{\mathrm{MA}}) \geq 0$. If $\nu(\mathcal{S}_{\mathcal{L}} \setminus \mathcal{I}_E^{\mathrm{MA}}) > 0$, the weighted average over correct studies converges to θ_0 by the law of large numbers applied to the restricted measure $\nu(\cdot \mid \mathcal{S}_{\mathcal{L}} \setminus \mathcal{I}_E^{\mathrm{MA}})$. □

Remark 7.3. Theorem 7.2 shows that asymptotic consistency is achievable despite population-level half-incorrectness, but only when the weight function correctly identifies and down-weights biased studies. This requires auxiliary information about study quality—a form of prior knowledge—which brings us naturally to the Bayesian framework.

8 Bayesian Hierarchical Meta-Analysis

8.1 The Prior Over Study Correctness

Following Section 10 of Ghosh (2026), the measure ν may be interpreted as a prior distribution over studies. The Half-Incorrectness Axiom then states

$$\mathbb{P}_{s\sim\nu}(s \in \mathcal{I}_E^{\mathrm{MA}}) \geq \tfrac{1}{2},$$

which is a prior constraint on the study population.

Definition 8.1 (Hierarchical Meta-Analytic Model)**.** The Bayesian hierarchical meta-analytic model is:

$$\begin{aligned}
\theta_0 &\sim \pi_0(\cdot), \\
\tau^2 &\sim \pi_\tau(\cdot), \\
\theta_i \mid \theta_0, \tau^2 &\sim \mathcal{N}(\theta_0, \tau^2), \\
\hat{\theta}_{s_i} \mid \theta_i, \sigma_{s_i}^2 &\sim \mathcal{N}(\theta_i, \sigma_{s_i}^2), \\
Z_i \mid \nu &\sim \mathrm{Bernoulli}(p_i), \quad p_i = \nu(\mathcal{I}_E^{\mathrm{MA}}),
\end{aligned}$$

where $Z_i = 1$ indicates that study s_i is incorrect. An additional bias term $B_i \cdot Z_i$ is added to $\hat{\theta}_{s_i}$ for incorrect studies.

Proposition 8.2 (Bayesian Updating Can Overcome Half-Incorrectness)**.** *Let the prior satisfy the Half-Incorrectness constraint $p_i \geq \frac{1}{2}$. Upon observing auxiliary quality indicators $\mathbf{q} = (q_1, \ldots, q_k)$ (e.g., risk-of-bias assessments), the posterior*

$$\mathbb{P}(Z_i = 1 \mid \mathbf{q}, \hat{\theta}_{s_i})$$

may fall below $\frac{1}{2}$ for individual studies of high apparent quality.

Proof. This follows from Proposition 4 of Ghosh (2026) applied to the individual study incorrectness indicator Z_i. The prior $\mathbb{P}(Z_i = 1) = p_i \geq \frac{1}{2}$, but upon observing a quality indicator q_i that strongly predicts correctness, $\mathbb{P}(Z_i = 1 \mid q_i)$ may be driven toward zero. The existence of such an update follows from Bayes' theorem and the positivity of the likelihood ratio $P(q_i \mid Z_i = 0)/P(q_i \mid Z_i = 1)$. □

Remark 8.3. Proposition 8.2 is the meta-analytic counterpart to Remark 2 of Ghosh (2026): population-level half-incorrectness does not preclude individual high-quality studies from being correctly identified. This provides a rigorous foundation for risk-of-bias assessments in systematic reviews, which aim to identify the high-quality subset of the study population.

9 Publication Bias as Measure Distortion

9.1 The Publication Filter

Publication bias—the tendency for statistically significant or positive results to be published at higher rates—distorts the empirical study distribution relative to the true study distribution.

Definition 9.1 (Publication Filter)**.** A *publication filter* is a measurable function

$$f : \mathcal{S}_{\mathcal{L}} \to [0, 1],$$

where $f(s)$ is the probability that study s is published. The *publication-distorted measure* is

$$\nu_f(A) = \frac{\int_A f(s)\, d\nu(s)}{\int f(s)\, d\nu(s)}, \qquad A \in \mathcal{A}.$$

Theorem 9.2 (Publication Bias Inflates Apparent Incorrectness)**.** *Suppose that $f(s)$ is negatively correlated with $|b(s)|$ under ν: that is, $\mathrm{Cov}_\nu(f(s), |b(s)|) < 0$. Then*

$$\nu_f(\mathcal{I}_E^{\mathrm{MA}}) \leq \nu(\mathcal{I}_E^{\mathrm{MA}}).$$

Conversely, if $f(s)$ is positively correlated with the magnitude of the reported estimate (favouring large effects regardless of validity),

$$\nu_f(\mathcal{I}_E^{\mathrm{MA}}) \geq \nu(\mathcal{I}_E^{\mathrm{MA}}).$$

Proof. From the definition of ν_f,

$$\nu_f(\mathcal{I}_E^{\mathrm{MA}}) = \frac{\mathbb{E}_\nu[f(s)\,\mathbf{1}_{s\in\mathcal{I}_E^{\mathrm{MA}}}]}{\mathbb{E}_\nu[f(s)]}.$$

When $\mathrm{Cov}_\nu(f, \mathbf{1}_{\mathcal{I}_E^{\mathrm{MA}}}) < 0$, the numerator is decreased relative to a neutral filter, giving $\nu_f(\mathcal{I}_E^{\mathrm{MA}}) < \nu(\mathcal{I}_E^{\mathrm{MA}})$. The reverse inequality follows symmetrically. □

Remark 9.3. Theorem 9.2 shows that publication bias can move the apparent incorrectness rate in either direction. Crucially, even if the published literature satisfies $\nu_f(\mathcal{I}_E^{\mathrm{MA}}) < \frac{1}{2}$, this is an artefact of the filter and not a true violation of Axiom 2.8 with respect to ν. Methods for correcting publication bias—such as trim-and-fill, PET-PEESE, or selection models [23, 22]—can be interpreted as attempts to recover ν from ν_f.

10 Entropy of Pooled Estimates

10.1 Differential Entropy of the Pooled Estimator

The companion paper Ghosh (2026) corrects a misinterpretation of binary entropy at the boundary $p = \frac{1}{2}$. An analogous correction applies to the meta-analytic setting.

Definition 10.1 (Differential Entropy of Pooled Estimate)**.** Let the pooled estimate $\hat\theta_{\Phi_w}$ have marginal density $p_\Phi(\cdot)$ under the study measure ν^k. The *differential entropy* of the pooled estimate is

$$h(\hat\theta_{\Phi_w}) = -\int p_\Phi(x)\log p_\Phi(x)\,dx.$$

Theorem 10.2 (Entropy Inflation under Heterogeneity)**.** *Suppose $\hat\theta_{\Phi_w}$ is approximately normally distributed with variance $V_\Phi = \sum_i w_i^2(\sigma_{s_i}^2 + \tau_\nu^2)$. Then*

$$h(\hat\theta_{\Phi_w}) = \tfrac{1}{2}\log(2\pi e\, V_\Phi).$$

As heterogeneity τ_ν^2 increases, $h(\hat\theta_{\Phi_w})$ increases, and the precision of the pooled estimate decreases correspondingly. The entropy of the pooled estimate is maximised (for fixed within-study variance) not at the boundary of any heterogeneity constraint, but in the limit $\tau_\nu^2 \to \infty$.

Proof. The entropy formula for a Gaussian $\mathcal{N}(\mu, V)$ is $h = \frac{1}{2}\log(2\pi e V)$, which is strictly increasing in V. Since V_Φ is strictly increasing in τ_ν^2, the result follows. □

Corollary 10.3 (Boundary Case and Maximum Uncertainty)**.** *At the boundary $\nu(\mathcal{I}_E^{\mathrm{MA}}) = \frac{1}{2}$, the binary entropy of study correctness is maximised, as established in Theorem 3 of Ghosh (2026). This boundary corresponds to maximal uncertainty about which studies to include in a meta-analysis—not to a minimally informative regime. Any meta-analyst who treats the $\frac{1}{2}$ boundary as "barely uncertain" commits an entropy inversion error.*

11 Network Meta-Analysis

11.1 Extension to Multiple Treatments

Network meta-analysis (NMA) extends pairwise meta-analysis to a graph of treatment comparisons [18]. The Ghosh (2026) framework extends naturally.

Definition 11.1 (Treatment Network)**.** A *treatment network* is a directed graph $G = (V, E)$ where each node $v \in V$ is a treatment and each edge $(u, v) \in E$ represents a comparison for which at least one study in $\mathcal{S}_\mathcal{L}$ provides evidence.

Definition 11.2 (Network Study Space)**.** For a treatment network G, the *network study space* is

$$\mathcal{S}_\mathcal{L}^G = \bigsqcup_{(u,v)\in E} \mathcal{S}_\mathcal{L}^{uv},$$

the disjoint union of study spaces for each comparison. Each $\mathcal{S}_\mathcal{L}^{uv}$ carries its own evaluation regime E_{MA}^{uv} and incorrectness set $\mathcal{I}_E^{uv}$.

Theorem 11.3 (Network Half-Incorrectness Propagation)**.** *Suppose Axiom 2.8 holds for each comparison-specific study space $\mathcal{S}_{\mathcal{L}}^{uv}$. Then the product measure $\nu^G = \bigotimes_{(u,v)\in E} \nu^{uv}$ on the network study space satisfies*

$$\nu^G\left(\mathcal{I}_E^G\right) \geq 1 - \prod_{(u,v)\in E} \left(1 - \nu^{uv}(\mathcal{I}_E^{uv})\right) \geq 1 - 2^{-|E|}.$$

Proof. The network incorrectness set is $\mathcal{I}_E^G = \bigcup_{(u,v)} \mathcal{I}_E^{uv}$ (at least one comparison-arm has an incorrect study pool). By inclusion-exclusion and the independence of comparison arms under the product measure,

$$\nu^G\left(\mathcal{I}_E^G\right) = 1 - \prod_{(u,v)} \left(1 - \nu^{uv}(\mathcal{I}_E^{uv})\right).$$

Since $\nu^{uv}(\mathcal{I}_E^{uv}) \geq \frac{1}{2}$ for each (u,v), we have $1 - \nu^{uv}(\mathcal{I}_E^{uv}) \leq \frac{1}{2}$, giving the stated bound. □

Remark 11.4. Theorem 11.3 shows that incorrectness propagates super-linearly in network meta-analysis: even if each comparison is only half-incorrect, a network of more than two comparisons has incorrectness probability greater than $\frac{3}{4}$. This formalises the practitioners' concern that NMA conflates direct and indirect evidence [13].

12 Connections Across Fields

Table 1 summarises how the framework developed here connects to core concepts across the disciplines that both produce and consume meta-analyses.

Field	Relevant Concept	Interpretation in This Framework
Evidence-based medicine	Systematic review quality	Epistemic Weight Function operationalises study quality; risk-of-bias tools approximate ω
Statistics	Model misspecification	Incorrect studies correspond to misspecified likelihood models; $B > \varepsilon$ measures degree of misspecification
Philosophy of science	Underdetermination	Multiple incompatible study designs may be mutually incompatible under E_{MA}
Econometrics	Specification search	Publication filters distort ν toward specificationally favourable studies
Psychology	Replication crisis	The replication failure rate empirically instantiates Axiom 2.8
Bayesian epistemology	Coherence and updating	Prior half-incorrectness does not preclude posterior correctness of individual studies
Network science	Graph-level inference	Network half-incorrectness propagates super-linearly with edge count
Information theory	Differential entropy	Heterogeneity inflates the entropy of the pooled estimate

Table 1: Cross-disciplinary interpretations of the measure-theoretic meta-analysis framework.

13 Relation to Existing Meta-Analytic Theory

13.1 Classical Frameworks

The classical theory of meta-analysis [9, 6] treats study incorrectness as an exception to be handled by sensitivity analysis, not a structural feature of the study population. The present framework inverts this assumption: incorrectness is the default prior, and correctness must be earned through the Epistemic Weight Function.

13.2 Heterogeneity Testing

The Q-statistic tests the null hypothesis $\tau^2 = 0$. In our framework, a significant Q is evidence that the effect distribution ρ is not degenerate, hence $\tau_\nu^2 > 0$ and Theorem 5.3 applies. The I^2 statistic [11], defined as the proportion of total variation attributable to between-study heterogeneity, can be interpreted as a continuous analogue of $\nu(\mathcal{I}_E^{\mathrm{MA}})$: both measure the degree to which the study population fails to support a universal pooled conclusion.

13.3 Robust Meta-Analysis

Methods such as three-level meta-analysis [5] and robust variance estimation [10] attempt to accommodate heterogeneity within the pooling paradigm. The present framework suggests a limit to such accommodation: when τ_ν^2 is large and $\nu(\mathcal{I}_E^{\mathrm{MA}}) \geq \frac{1}{2}$, no within-paradigm robustification can reduce the pooling bias bound established in Theorem 4.3.

14 Limitations

The framework developed here inherits the limitations identified in Section 15 of Ghosh (2026), and introduces several of its own.

First, the admissible study space $\mathcal{S}_\mathcal{L}$ depends on the language $\mathcal{L}$ and the evaluation regime E_{MA}. Different PICO (Population, Intervention, Comparator, Outcome) specifications yield different admissible spaces and potentially different incorrectness rates.

Second, the study measure ν is not canonical. In practice, a systematic review works with a convenience sample from the published literature, which is ν_f rather than ν. Correcting for publication bias is equivalent to estimating ν from ν_f, an ill-posed inverse problem.

Third, the bias threshold ε is chosen by the analyst. Different choices of ε yield different incorrectness sets and different bounds. Sensitivity analysis over ε is therefore essential.

Fourth, the lower bound in Theorem 4.3 assumes $B > \varepsilon$. When incorrect studies have small bias ($B \approx \varepsilon$), the bound is vacuous. The framework provides the most traction in settings where incorrect studies produce non-trivial bias.

Fifth, Theorem 7.2 requires the Epistemic Weight Function to correctly identify incorrect studies. In practice, this identification is imperfect, and the weight function will itself be subject to uncertainty—a second-order application of the Half-Incorrectness Axiom.

15 Conclusion

This paper has developed a rigorous measure-theoretic foundation for meta-analysis within the Half-Incorrectness framework of Ghosh (2026). The central contributions are as follows.

The meta-analytic study population is formalised as a measurable space $(\mathcal{S}_\mathcal{L}, \mathcal{A}, \nu)$ equipped with a regime-relative incorrectness set $\mathcal{I}_E^{\mathrm{MA}}$. The Half-Incorrectness Axiom, $\nu(\mathcal{I}_E^{\mathrm{MA}}) \geq \frac{1}{2}$, is instantiated as a structural postulate about the study population, consistent with empirical findings from large-scale replication efforts.

From this axiom, we have derived:

(i) a non-trivial lower bound on the expected pooling bias under uniform weighting (Theorem 4.3);

(ii) the incoherence of universal pooling under heterogeneity above a critical threshold (Theorem 5.3, Corollary 5.4);

(iii) reflexive non-exemption of the meta-analytic protocol (Theorem 6.2);

(iv) coherent self-failure without contradiction (Proposition 6.3);

(v) asymptotic consistency under the Epistemic Weight Function (Theorem 7.2);

(vi) the role of publication filters as measure distortions (Theorem 9.2);

(vii) super-linear propagation of incorrectness in network meta-analysis (Theorem 11.3);

(viii) the inflation of pooled-estimate entropy under heterogeneity, and the correction of the boundary-case entropy interpretation (Theorem 10.2, Corollary 10.3).

The framework does not counsel pessimism about meta-analysis. It specifies the conditions under which meta-analysis succeeds—when an Epistemic Weight Function correctly identifies the correct studies—and the conditions under which it fails—when heterogeneity is high, the study measure is severely distorted, or the meta-analytic protocol itself is reflexively incorrect. The rigorous formulation of these conditions is the prerequisite for principled evidence synthesis.

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